

UNIT 1

INTRODUCTION TO DIGITAL IMAGE

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1.1 INTRODUCTION

There is a famous saying that a picture is equivalent to thousands of words, and this turns out to be true when we come to the field of image processing and pattern recognition. It is the interpretation of image which determines various types of outcomes viz. for an arts person it may carry a different meaning but for a science person the meaning of that image may be entirely different. Based on the utility of the images, different formats are devised to store the different type of images, which are stored in different file formats and they contain a variety of information in the form of colour, contrast, intensity, brightness etc., but resolution of these images holds a great importance in the field of image processing, because it affects the quality of image when it is scaled up. This unit relates to the understanding of essentials of image processing and to find major applications of image processing in our everyday life. Broadly speaking, image processing is an area that deals with manipulation of visual information called images, and one of the major objectives of image processing is to improve the quality of pictorial information for better human interpretation and to facilitate the automatic machine interpretation of images.

We shall begin our discussion by defining an image in Sec. 1.2 and will continue image acquisition and image digitization in Sec. 1.3 and Sec. 1.4 respectively. In the subsequent sections, we shall define the basic terms and concepts, which will be used throughout the course such as representation of image, resolution of image, characteristics of image and types of images. We shall end the unit by discussing various areas in which digital image processing is used.

Now, we shall list the objectives of this unit. After going through the unit, please read this list again and make sure that you have achieved the objectives.

Objectives

After going through this unit, you should be able to

- define and represent an image;
- define the various terms used in digital image processing;
- apply sampling and quantization for image digitization;
- list various classifications of images along with their descriptions;
- relate the areas where digital image processing can be applied.

Now, let us find the answer to the question “what an image is?” in the following section.

1.2 INTRODUCTION TO AN IMAGE?

When we studied optics as a subject in the field of physics, we learned that when two or more rays of light meet at a point an image is formed, and we studied various phenomena like reflection, refraction, diffraction, dispersion, scattering etc., this means we were in the process of image generation and analysis, even much before the development of computers. But, with the advent of the computers in our day to day life this field of image processing has transformed into an entirely new field i.e. digital image processing, where sampling and quantization techniques are applied to convert the image into its discrete form, this process of conversion is collectively known as image digitization, after digitization the image is readily available in a form suitable for further processing by digital computers. We shall discuss about the concept of sampling and quantization in the subsequent sections, here we will discuss about the concept of image only.

Many times the term ‘picture’ is used, but this term relates to the analog or raw image data and the term ‘image’ is used to refer to digital data that is suitable for the processing of images using digital computers. Infact Images are imitations of a real world objects. Image may be considered as a projection of the real world (to be more precise 2D projection of 3D world). From a photographers point of view it is a photograph (i.e. projection of real world), and from the point of view of a computer engineer an image may be a two-dimensional (2D) signal. Therefore, an image is a two-dimensional function $f(x, y)$ where for

each position (x, y) in the projection plane, the values of the function $f(x, y)$ represent the amplitude or light intensity of the image.

Images are of two types, analog and digital. Analog image can be mathematically represented as a continuous range of values representing position and intensity. A digital image is composed of picture elements called **pixels**.

You may note here that through out the course we shall use the word image which refers to digital image.

As we mentioned earlier an image is a projection of the environment under consideration, if the environment is 3D then its projection will be in 2D. If we are having an n -Dimensional space then its projection will be of $n-1$ dimensions. Magnetic resonance images and computerized tomography (CT) images, which are 3D images, are mathematically represented by 3D function $f(x, y, z)$, where x, y, z are spatial coordinates which may represent the amplitude or intensity or any other parameter of the image.

A digital image is defined as a 2D discrete signal that varies over the spatial coordinates x and y , and can be written mathematically as $f(x, y)$. It is also an $n \times n$ array of elements, and each element represents the sampled intensity.

In general, the image $f(x, y)$ is divided into X rows and Y columns. The coordinate ranges are $\{X = 0, 1, \dots, m-1\}$ and $\{Y = 0, 1, 2, \dots, n-1\}$. The image can be written as a mathematical function $f(x, y)$ as given below:

$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & f(0,2) & \dots & f(0,y-1) \\ f(1,0) & f(1,1) & f(1,2) & \dots & f(1,y-1) \\ \dots & \dots & \dots & \dots & \dots \\ f(m-1,0) & f(m-1,1) & f(m-1,2) & \dots & f(m-1,n-1) \end{bmatrix}$$

A sample digital image is shown in Fig. 1(a) and its equivalent matrix is shown in Fig.1(b).

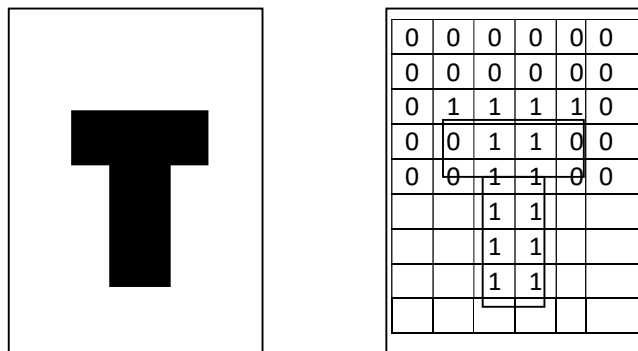


Fig. 1: (a) Sample Digital Image and (b) Matrix of the Image

In Fig. 1(b), the elements are shown as either 0 or 1, which represents the value of the function $f(x, y)$, each element is known as **pixels** i.e. 'picture element'. They represent the discrete data of any digital image, and serve as the actual building block of digital images. The value of the function $f(x, y)$ at every point indexed by a row and a column is a number and has no units, and it is known as the **grey value** of intensity of the image at that point.

You may again refer to Fig. 1(b), the number of rows in a digital image is called **vertical resolution**. The number of columns is called **horizontal resolution**. We shall discuss about these concepts of image resolution, later in this unit. The number of rows and columns describes the **dimensions** of the image. The image size is often expressed in terms of the rectangular pixel dimensions of the array. Images can be of various sizes. Some examples of image size are 256 x 256, 512 x 512, etc. For a digital camera, the image size is defined as the total number of pixels (specified in megapixels). For example an image with resolution 2048x2048 will have 4×10^6 pixels, that is, 4 Megapixels. Millions of pixels combine together to give a digital image, and their meaning varies with context i.e. a pixel can be considered a single sensor, photosite (physical element of the sensor array of a digital camera), element of a matrix, or display element on a monitor.

Generally, the value of the pixel is the intensity value of the image at that point, which is quantized value of the light that is captured by the sensor at the point. Typically the quantization is done in 256 levels. Thus the pixels will have values going from 0 to 255, and every pixel will need 8bits to store this information. But many times, the value of the pixel is not always the intensity value. For example, in the case of an X-ray image, the value of the pixel indicates the attenuation of the X-ray at the point. Similarly, the average Magnetic Resonance (MR) signal intensity denotes the pixel value in an MRI.

You may try the following exercises.

-
- E1) What do you mean by the term image file format? Mention some of the frequently used image file formats.
 - E2) Differentiate analog image and digital image. Also, give example in support of your answer.
-

Now, in the following section, we shall discuss image acquisition.

1.3 IMAGE ACQUISITION

According to the fundamental concept of optics, when two or more rays of light meet at a point due to reflection or refraction or any other phenomenon, an image is formed. The information of happened phenomenon is recorded or acquired by the respective sensors as an image of that event. The process of capturing real world images and storing them into a computer is called **image acquisition**.

The question arises here, that how do we get the image in digital form. The answer is that the sensors are actually the electronic devices which taps the various types of signals, like thermal or optical or electromagnetic or any other type. Generally, the data is gathered in the analog form, which is digitized through a digitizer, and this digital signal is finally utilized by the digital computer for its processing. This leads to three types of image processing :—

- Optical Image Processing
- Analog Image Processing
- Digital Image Processing

We define these types briefly:

- **Optical Image Processing:** An optical image of 3D object is in fact its 2D projection, which is the continuous distribution of light on a 2D surface. This 2D projection holds various types of information like 3D objects that are in focus. It also includes the study of the radiation source, and other optical processes. This optical image is available in the optical form until it is converted into analog form, leading to the area of analog image processing.
- **Analog Image Processing:** This relates to the processing of analog signals using analog circuits. Analog signals are the time varying continuous signals, often referred to as pictures. These analog signals are transformed into digital image through the process of sampling and quantization, performed by using a digitizer and the process is known as digitization. Fig. 2 shows the steps of analog image processing.



Fig. 2: Analog Image Processing

You may note that the imaging systems that use film for recording images are also known as analog imaging systems, in medical imaging, still films are used, because these films provide better quality than digital systems.

- **Digital Image Processing:** It relates to the field of using digital circuits, systems, and software algorithms to carry out the image processing operations, which include quality enhancement of an image, counting of objects, image analysis, and many more.

Now, we know that when an interaction happens between object and rays of light (or signals) some optical or thermal or relevant phenomenon happens. The phenomenon is recorded or acquired by the respective sensors as an image of that event. This image relates to the combination of millions of pixels, each of which is carrying the information for function $f(x, y)$ or $f(x, y, z)$, which may be intensity or X-ray attenuation value, or Magnetic Resonance Intensity or any other parameter. This is gathered by using respective sensors or other

devices, this data gathering relates to the field of image acquisition, which is actually the first phase of Digital Imaging. This acquired information is stored in to the memory of the digital computer for necessary processing, the final outcome of processed image is transmitted to the concerned as an information. This means that the relevant tasks performed by any digital imaging system are **data acquisition, storage, manipulation, and transmission**.

Thus, a typical digital imaging system or digital imaging workstation involves components for the image acquisition, storage, processing, and transmission. Types of imaging systems and acquisition systems are shown in Fig. 3.

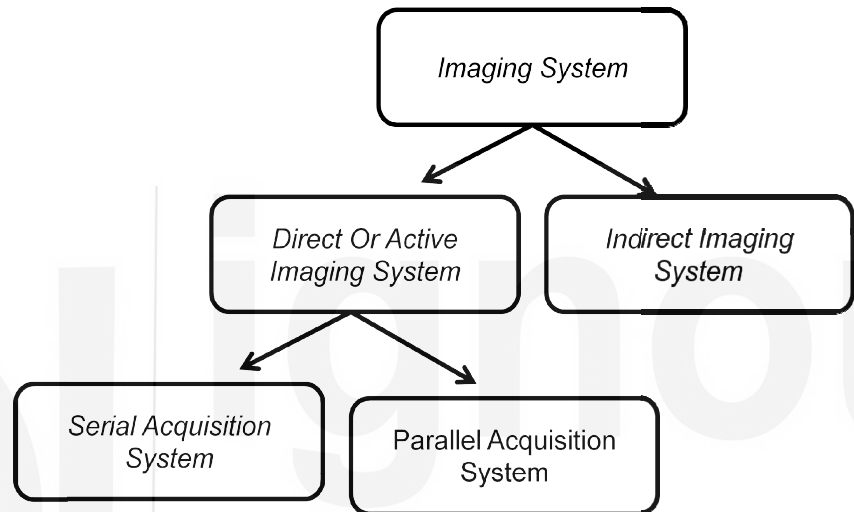


Fig. 3: Types of Imaging and Data Acquisition Systems

Let us now discuss these types briefly.

Direct or Active Imaging Systems: These are the systems where no temporary storage on film or chemical processes is required. In such systems the acquired images are in a recognizable form. We can say that in such systems the final image is a digital image. For example, human eye, digital camera, scanning confocal microscopes, etc.

The direct or active imaging systems are divided in to two sub categories i.e. serial and parallel acquisition systems. The human eye and the digital camera are example of parallel active imaging systems, where as scanning microdensitometer and scanning confocal microscopes are examples of serial acquisition imaging systems. The advantage of direct imaging systems is that the final image is a digital image.

Indirect Imaging Systems: In comparison to Direct imaging systems, the Indirect imaging systems involves data processing or reconstruction before producing an image for observation. In indirect imaging systems, the image is stored in a film, which is rendered observable by a chemical process. Therefore, there is a delay in the production of images from films.

You may now try the following exercises.

E3) Distinguish among optical image processing, analog image processing and digital image processing using examples.

E4) Differentiate between direct and indirect imaging system.

After data or image acquisition, image digitization is to be performed. We shall be using the image digitization process in image digitization.

1.4 DIGITIZATION OF IMAGES

In this section, we will try to find out the answers to three basic questions. The first question is why do we need digitization? Then we will try to find the answer to what is meant by digitization and thirdly, we will go to how to digitize an image.

We have learned from the previous sections that images are actually the imitations of real-world objects, which is generally a two-dimensional (2D) signal $f(x, y)$, where the values of the function $f(x, y)$ represent the amplitude or intensity at different points (pixels) of an image. In the processing of the analog signals or information by using digital computers, we need to convert this analog image into a discrete form using the process of digitization. This process of Digitization involves subprocess of sampling and quantization, performed through a Digitizer. The process of sampling and quantization transforms the acquired data in to the form which is suitable for further processing by digital computers.

Thus, image digitization refers to the process of sampling and quantization of the analog signals, which is required to convert the analog signal in to digital form. The digital image processing is performed using Digitizers, and this digitization process consists of two steps. One is called **sampling** and the other is called **quantization**. Sampling refers to considering the image only at a finite number of points, each image sample is called a pixel and Quantization refers to the representation of the grey level value at the sampling point using finite number of bits. For example 256 level quantization results in image storage of 8 bits per pixel.

Let us discuss the two concepts in detail.

Sampling

Consider any image. As you know that an image can be viewed as a 2 dimensional function given in the form of $f(x, y)$. Now, the image has certain length and certain height. Suppose the height of the image is H and the length of the image is L . The units of length and height would be the same. We can identify the coordinates X any Y at any space on the image. As you have seen the matrix representation of function

$f(x, y)$ in Sec. 1.2, x -axis is taken vertically and y -axis is taken horizontally, so X coordinate will vary from 0 to H and Y coordinate will vary from 0 to L . At any point (X, Y) we identify two features and write XY as product of them. Suppose these features are $r(X, Y)$ and $i(X, Y)$ represent the reflectance of the surface point and intensity of light respectively. If we consider that the maximum and minimum values of intensity are $I_{\max}$ and $I_{\min}$, then in case of continuous image $f(x, y)$ cannot attain the values either according to XY or $r(X, Y).i(x, y)$.

From the theory of real numbers you know that given any 2 points, there are infinite numbers of points. So again, when I come to this image as X varies from 0 to H , there can be infinite possible values of X between 0 and H . Similarly, there can be infinite values of Y between 0 and L . So effectively, that means that if we want to represent this image in a computer, then this image has to be represented by infinite number of points and secondly when we consider the intensity value at a particular point, it varies between certain minimum $I_{\min}$ and certain maximum $I_{\max}$ values, which is infinite in number.

It is clear from here that the number of points in any case would be infinite which would not be possible for computer to work on. Therefore, a way to overcome from this problem is to consider some discrete set of points, and this process of taking discrete data for any image is known as sampling.

From this discussion, we can say that since the computers can not handle the continuous data, so the continuous signal needs to be converted into digital signal and the process of this conversion is referred to as sampling.

We can visualize the process as overlaying an uniform grid on the image and sampling the image function at the center of each grid square. As we make the grid finer, we get better resolution of the image, but as we make the grid coarser, we observe more "*pixelization*". At each pixel (or at each grid square) we usually represent the gray level value using an integer ranging from 0 for black to 255 for fully white.

To sample the signal, the signal should be frozen, as sampling cannot be applied to moving signals. If the signal is frozen for T seconds, T is called the sampling period. The **sampling period** is measured in seconds, milliseconds, or microseconds. For example, if the sampling rate is 1000Hz, it means that the signal is sampled every millisecond.

The **sampling rate** or **sampling frequency** is the reciprocal of the sampling period and is measured in samples per second or Hertz. The sampling process is the multiplication of a continuous signal $f(t)$ by a railing function $r(t)$. Railing function can be a pulse of unit amplitude at exactly the sampling time instant. We can also say that If the signal is frozen for T seconds (T is the sampling period) then sampled function $f(n)$ is mathematically represented as the product of continuous signal $f(t)$ by a railing function $r(t)$,

$$f(n) = f(t) \times r(t) = f(nT)$$

All these components are graphically shown in Fig. 4.

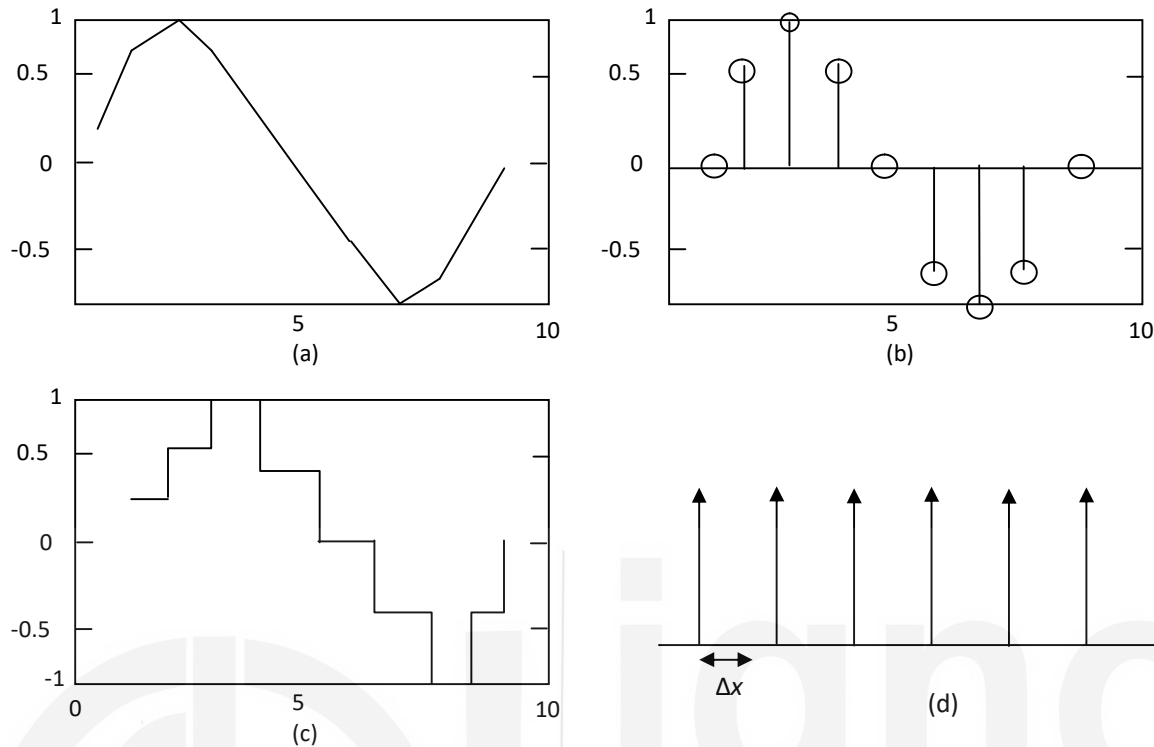


Fig. 4: Sampling process (a) Original signal $f(t)$ (b) Sampled image $f(n) = f(t) * r(t)$ (c) Reconstructed image (d) Train of the impulse function $r(t)$

Since, sampling is a reversible process, the reconstruction of original signal from the sampled signal is possible in both frequency and time domain. In frequency domain, the information of the base band component is required, and this base-band component over the band $\pm f_s$ (where f_s is the sampling frequency) can be extracted from the infinite spectrum of the infinite spectrum of the sampled signal, by using the low pass filters. (This statement is derived from the Shannon's Sampling Theorem). In image processing, the sampling frequency decides the distance between the samples. This distance determines the linear pixel size. Alternatively, the equivalent operation in time domain is the interpolation operation where an ideal low-pass filter is used in convolution with the interpolation function (e.g., sine function). The function $f(t)$, the railing function $r(t)$, and the sampled and reconstructed functions are all shown in Fig. 5.

This idea of one-dimensional sampling can be extended to 2D images also. The 2D railing function is known as **comb function**. It is arranged as a rectangular grid of unit impulses separated by Δx and Δy . The 2D sampling process can again be viewed as the multiplication of the railing function with the continuous function to give discrete samples. The values of Δx and Δy play an important role in image processing. Normally they are kept the same in both the horizontal and the vertical directions, so that the pixels will be a square pixels. This is called **pixel aspect ratio**. In addition, the size of the pixel is important for image quality. If the size is very large, there will be a lesser number of pixels. Hence, the details become less, which makes the image distracting and

meaningless. This is called **pixelization error** where the grey level discontinuities at the edges of the pixel become poor. Then question arises that, What should be the ideal size of the pixel? Should it be big or small? The answer is give by the Shannon-Nyquist theorem.

As per Shannon-Nyquist theorem, the sampling frequency should be greater than or equal to $2 \times f_{\max}$, where $f_{\max}$ is the highest frequency present in the image. Otherwise, the original signal cannot be reconstructed. In other words, the number of samples required is dictated by Shannon-Nyquist theorem, this sampling theorem can be stated in terms of distance d as $d \leq \frac{1}{2f_{\max}}$. Alternatively, it should be less than or equal to the smallest detail that is present in the image.

Another frequency that is helpful in image processing is the Nyquist frequency. The Nyquist frequency (f_N) is $1/2 \times$ (sampling frequency). Therefore, the Nyquist frequency (f_N) should be greater than or equal to $f_{\max}$. If the sampled image has frequencies higher than the Nyquist frequency, it results in a condition called aliasing where the high frequencies masquerade as low frequency components. This results in an image where interpretation becomes difficult. This problem is called **aliasing**.

Now, let us solve the following examples to understand this concept better.

Example 1: What should be the physical size of a 2D image of a document with dimensions is 2400×2400 , when scanned at 300dpi. Here dpi stands for dots per inch.

Solution: The physical size of image is

$$\begin{aligned} &= \frac{\text{Number of pixels in width}}{\text{Resolution}} \times \frac{\text{Number of pixels in height}}{\text{Resolution}} \\ &= \frac{2400}{300} \times \frac{2400}{300} \\ &= 8 \text{ inches} \times 8 \text{ inches} \end{aligned}$$

Example 2: If the physical size of a medical image is 8×8 inches and the sampling resolution is 5 cycles/mm, then how many pixels per cycle are required to have a better quality image? Will an image of size 256×256 be enough?

Solution: It is given that the sampling resolution is 5 cycles/mm. Therefore, for better quality, 2 pixels per cycle are required. This is because sampling theorem states that the sampling frequency should be greater than twice the maximum signal frequency. This means that 10 pixels per mm are required. So, the pixel size is 0.1 mm.

You may note here that this double rates in both directions are called Nyquist rates.

The given image size (since 1 inch = 2.54 cm) is

$$= 8 \times 2.54 \times 8 \times 2.54 \text{ cm}^2$$

$$= 20.32 \times 20.32 \text{ cm}^2$$

Therefore, the minimum number of required pixels = 2032×2032 pixels.
So, an image of size 256×256 is not enough.

Now try the following exercises.

-
- E5) The dimension of an image is 5 x 8 inches and the frequency is 500 dots per inches in each direction. Find the number of samples required to preserve the information in the image.
-

Now let us discuss the other step of image digitization, which is image quantization.

Quantization

Image quantization is the process of converting the sampled analog value of the function $f(x, y)$ into a discrete-valued integer. An analog signal has an infinitely larger number of distinct values. So, it is necessary to convert the continuous values into a smaller set through the process of quantization. The sampled analog image i.e. a natural image has continuously varying shades and colours and is known as a continuous tone image. This continuous tone image is required to be converted into a discrete image. The image quantizer maps a continuous value x into a discrete variable x' , where discrete points of grey-tone or brightness are used.

The process of quantization involves the partitioning of input values into equally spaced intervals. The end points of the interval are called **decision boundaries**. Let the decision boundaries be given by $B = \{b_0, b_1, \dots, b_m\}$, and let the input values be in the range $-X_{\max}$ to $+X_{\max}$. The length of the interval between successive decision

boundaries i.e. the step size (Δ) is given by $\frac{2X_{\max}}{m}$. The midpoint

between successive decision boundaries is called **output or reconstruction level** and is given by $(R) = \{y_1, y_2, \dots, y_m\}$. Then

quantization is in the range $\left[\text{from } -\frac{\Delta}{2} \text{ to } \frac{\Delta}{2} \right]$, and the number of bits

necessary to encode the output levels is given by $R = \lceil \log_2 m \rceil$.

Try the following exercises.

-
- E6) What is the role of Sampling and Quantization in the process of Digitization?
- E7) What do you mean by pixel aspect ratio and pixelization error?

- E8) How Shannon-Nyquist theorem relates to the determination of the ideal size of the pixel?

In the following section, we shall discuss the ways how an image is represented.

1.5 REPRESENTATION OF DIGITAL IMAGE

In conventional image processing, a matrix is used to represent intensity values. Many times we need to convert this matrix to a vector and then use the normal equations or regularization based methods to represent intensity values. The normal matrix representation of image is called the Spatial Domain. There are other representation of images obtained by transforming the spatial domain image like Fast Fourier Transform (FFT), Discrete Cosine Transform (DCT), Discrete Wavelet Transform (DWT) etc. Each of these transformations represents the image in the form of a mathematical formula. Further, some Partial Differential Equations (PDE) based methods use kernels to process images using 2D convolution. Sometimes we use the notion of images as point clouds and then construct a graph based on some distance measure. Then we study the Laplacians (Diffusion maps) or the Hamiltonian operators of these graphs. The Eigen values of these operators lead to the notion of heat kernel signatures and wave kernel signatures. We have these alternate views of an image based on the application, we will study some of these image representation mechanism in this section.

Going back to the basics, a digital image represents a mapping on a visual scene recorded by an optical sensor into an array of picture element intensities. Here we are discussing some of the models of image representation.

- **Pixel model:** In this model, the RGB optical sensors maps to a vector of discrete intensities which are integers. Let $p: R \times G \times B \rightarrow Z \times Z \times Z$ defined by $p(r, g, b) = (zR, zG, zB)$, where r, g, b are real valued colour intensities and zR, zG, zB are integer-valued colour intensities in the range from 0 to 255. In this case, a pixel is represented by a 1×3 column vector used to "paint" a tiny square in an image display.
- **Vector model:** In this model, the RGB optical sensor values are mapped to vectors and the displayed image conforms to a form of triangulated image space instead of a collection of pixels. In this model, a vector image is a collection of vertices and edges. The end result of vector images is the formation of a basis for applications such as Adobe Photoshop and various video games.

You can try the following exercise.

-
- E9) Compare Pixel Model and Vector Model for the representation of Digital Image.
-

In the following section we will discuss about the various types of images, and their respective characteristics.

1.6 TYPES OF IMAGES

Classification of images can be performed on the basis of various criteria like attributes, colour, dimension and data types. Based on Attributes criteria the images are classified as raster and vector, whereas on the basis of colour we can classify an image as binary, greyscale, true colour and pseudo colour. Further on the basis of dimensions, we have 2D images and 3D images, and on the basis of data types the images are classified as signed integer type, unsigned integer type, float type, logical and double type. So there is no single accepted way of classifying images. The image classification is described in detail now:

- **Classification of Images on the basis of Attributes:** Based on attributes of any image, it is classified as raster and vector graphic images.
 - **Vector graphics** uses graphic primitives (point/line/circle/ellipse etc) to describe an image. Hence, the notion of resolution is practically not present in graphic.
 - **Raster images** are pixel-based, and hence their quality is dependent on the number of pixels. So, operations such as enlarging or blowing-up of a raster image often result in quality reduction.
- **Classification of images on the basis of colour:** On the basis of colour, the images can be classified into the following categories:
 - **Monochrome images** are the images where the colour component is absent, they are further classified as Grey scale and binary images.
 - **Grey scale images** : The term grey scale refers to the range of shades between white and black or vice versa, such images have many shades of grey, and eight bits ($2^8 = 256$) are enough to represent the grey scale because the human visual system can not distinguish more than 32 different grey levels, and the additional bits are necessary to cover noise margins. Most medical images such as X-rays, CT images, MRIs, and ultrasound images are grey scale images.
 - **Binary images** : The binary images are just special case of grey scale images, where the process of Thresholding is applied, they are actually Bi-level images where the pixels assumes the values of 0 or 1, i.e. only one bit is sufficient to represent the pixel value. The thresholding process involves the comparison of the threshold value i.e. the pixel value is compared with the threshold value, if the pixel value of the grey scale image is greater than the threshold value, the pixel value in the binary

image is considered as 1, Otherwise, the pixel value is 0. So far as the utility of Binary images is concerned, they are often used in representing basic shapes and line drawings. They are also used as masks. In addition, image processing operations produce binary images at intermediate stages.

- **True colour (or full colour)** images are the images where the pixel has a colour that is obtained by mixing the primary colours red, green, and blue. Each colour component is represented like a grey scale image using eight bits. Mostly, true colour images use 24 bits to represent all the colour. Hence true colour image can be considered as three-band images. The number of colours that is possible is 256^3 (i.e. $256 \times 256 \times 256 = 1,67,77,216$ colours). A display controller then uses a digital-to-analog converter (DAC) to convert the colour value to the pixel intensity of the monitor, refer to Fig.5 .



Fig. 5: True Colour Image

True colour images carry the full range of available colours in themselves. Thus, they are quite similar to the actual object and hence called true colour images. Further, the true colour images do not use any lookup table but store the pixel information with full precision.

- **Pseudocolour images** are in fact false colour images, their colour component is manipulated artificially, based on the interpretation of data. Like true colour images, pseudocolour images are also used widely in image processing. We know that true colour images are called three-band images. But, in remote sensing application, multi-band images or multi-spectral images are generally used. These images, which are captured by satellites, contain many bands. A typical remote sensing image may have 3 to 11 bands in an image. This information is beyond the human perceptual range. Hence it is mostly not visible to the human observer. So colour is artificially added to these bands, so as to distinguish the bands and to increase operational convenience. These are called artificial colour or pseudocolour images. Pseudocolour images are popular in the medical domain also. For example, the doppler colour image is a pseudocolour image.

We shall discuss few examples for the better understanding of the topic.

Example3: Determine the storage space required to store 1024 x 1024 pixels of a binary image?

Solution: For a binary image, one bit is sufficient for representing the pixel value. So, the number of bits required will be $1024 \times 1024 \times 1 =$

$10,48,576 \text{ bits} = [(10,48,576)/8] \text{ bytes} = 1,31,072 \text{ bytes} = 128 \text{ KB}$. Since, 1 KB is 1024 bytes, the storage requirement is 128 KB.

Example 4: What is the storage requirement for a 1024 x 1024, 24-bit colour image?

Solution: Since colour images are three-band images (red, green, and blue components), the storage requirement is $1024 \times 1024 \times 3 \text{ bytes} = 31,45,728 \text{ bytes}$. Since 1 KB is 1024 bytes, the storage requirement is 3072 KB or 3 Mega pixels.

We shall resume the discussion on image classification.

- **Classification of Images on the basis of Dimensions :** Images can be classified on the basis of dimensions also. Normally, digital images are a 2D rectangular array of pixels. If another dimension, of depth or any other characteristic, is considered, which may be necessary to use, then a higher-order stack of images like 3D images are produced. A good example of a 3D image is a volume image, where pixels are called voxels. By '3D image', it is meant that the dimension of the target in the imaging system (may be a scene or an object.) is three dimensional (x, y, depth). In medical imaging, some of the frequently encountered 3D images are CT images, MRIs, and microscopy images. These are stored basically as 2D image slices taken across the body or the skull. Range images (often used in remote sensing application) are also 3D images as they also incorporate the depth information.
- **Classification of Images on the basis of Data Types:** Images may be classified based on their data type. Sometimes, image processing operations produce images with negative number, decimal fraction, and complex number. For example, Fourier transforms produce image involving complex numbers. To handle negative numbers, signed and unsigned integer types are used. In these data types, the first bit is used to encode whether the number is positive or negative. For example, the signed data type encodes the numbers from -128 to 127 where one bit is used to encode the sign. In general, an $n - 1$ bit signed integer can represent integers from -2^{n-1} to $2^{n-1} - 1$, a total of 2^n . Unsigned integers represent all integers from 0 to $2^n - 1$ with n bits.

Floating-point involves storing the data in scientific notation. For example, 1,230 can be represented as 0.123×10^4 , where 0.123 is called the significant and the power is called exponent. There are many floating-point conventions.

The quality of such data representation is characterized by parameters such as data accuracy and precision. Data accuracy is the property of how well the pixel values of an image are able to represent the physical properties of the object that is being imaged. Data accuracy is an important parameter, as the failure to capture the actual physical properties of the image leads to the loss of vital

information that can affect the quality of the application. While accuracy refers to the correctness of measurement, precision refers to the repeatability of the measurement. In other words, repeated measurements of the physical properties of the object should give the same result. Most software use the data type 'double' to maintain precision as well as accuracy.

Now try the following exercises.

E10) Make a list of all the types of classifications of images based on various parameters.

E11) Compare true Colour Images and Monochromatic Images.

In this section, we learned various types of images. Now it is time to learn about the fundamental characteristics of images because to improve the quality of an image, it is necessary to assess its quality, and this assessment requires understanding of characteristics of images, which is discussed in the following section.

1.7 IMAGE CHARACTERISTICS

To improve the quality of an image, it is necessary to assess its quality, and this assessment requires understanding of characteristics of images. Some of the essential characteristics of image are Intensity, Contrast, Brightness, Noise and Resolution, we will discuss them one by one.

- **Intensity:** The term Intensity refers to the amount of light or the numerical value of a pixel, it is the measure of energy of a wave, which is directly proportional to square of amplitude of the signal. From the point of view of image processing it is a numerical value which represents a pixel, for *example* for an 8-bit gray scale image the value of a pixel can be in the range of [0,255], where (say) numeric value "0" represents black colour and the numeric value "255" represents white colour. Therefore, *the higher the value of a pixel (i.e. intensity) whiter the pixel will appear*, i.e. in a grey scale images, intensity is depicted by the grey level value at each pixel i.e., two pixels with grey level value 127 and 220, then it can be interpreted that pixel with value 127 is darker than pixel with value 220 .
- **Contrast:** The term Contrast of an image relates to recording of the differences in the magnitude of the intensity at the surface of an object. There are many ways in which contrast can be measured, i.e. it can be described as a product of the sensor signal contrast (this depends on the energy source and the physical properties of the object) and detector contrast (this depends on the way signal is detected, captured, and stored). But, a common

measurer of contrast (C) involves intensity of foreground (I_{object}) and background ($I_{\text{background}}$) objects, i.e.

$$C = \frac{I_{\text{object}} - I_{\text{background}}}{I_{\text{object}} + I_{\text{background}}}$$

Where, I_{object} is the average pixel intensity of the object pixels and $I_{\text{background}}$ is the average pixel intensity of the background. Another useful contrast measure using the same parameters I_{object} and $I_{\text{background}}$ is

$$C = \log_{10} \frac{I_{\text{object}}}{I_{\text{background}}}$$

We know, Contrast of an image relates to recording of the differences in the magnitude of the intensity at the surface of an object. So, *Contrast is the difference between maximum and minimum pixel intensities in an image*. Consider two images A & B having pixel intensities between 30 to 200 and 70 to 130, respectively. Then A has more contrast than B. Again contrast is also relative. Contrast can be simply explained as the difference between maximum and minimum pixel intensity in an image. For example, consider an image, whose maximum and minimum value of intensity is 100 i.e. same then contrast will be zero because, Contrast = maximum pixel intensity – minimum pixel intensity = 100 – 100 = 0.

- **Brightness:** It refers to the average pixel intensity of the image, Similar to contrast, the image should be reasonably bright to show all the information to the viewer. However, excessive brightness may affect the quality of the image. An image may have a higher brightness but if the intensity is not optimal, the image again will not exhibit good quality. *Brightness resolution* is defined as the number of identified distinct colours and it is also called as *colour resolution*.

Brightness is a relative term which can be understood visually you can say the higher the intensity the brighter is the pixel. Since brightness is a relative term, so brightness can be defined as the amount of energy output by a source of light relative to the source we are comparing it to. In some cases we can easily say that the image is bright, and in some cases, it's not easy to perceive. Now, Question is How to make an image brighter? Brightness can be simply increased or decreased by simple addition or subtraction, to the pixel values of the image matrix. Consider a black image of 5 rows and 5 columns. All the entries of the image matrix are going to be zero since it is a completely black image. To make it brighter we shall perform some operation on this image. What we will do is, that we will simply add a value of 50 (or any value from 1 to 255) to each of the matrix value of image. After adding the image would become somewhat brighter than its previous state of brightness.

- **Noise:** Noise is an unwanted disturbance that causes fluctuations in the pixel value. It is a random or stochastic process and hence its true value cannot be predicted accurately. However, noise obeys all the statistical properties. So a pixel is characterized as a random

variable for statistical analysis. It is the next important characteristic, related to the quality of image, Image applications are frequently affected by the noise present in the image.

- **Resolution:** Resolution refers to the number of pixels in an image. Resolution is sometimes identified by the width and height of the image as well as the total number of pixels in the image. For example, an image that is 2048 pixels wide and 1536 pixels high (2048 x 1536) contains (multiply) 3,145,728 pixels (or 3.1 Megapixels). You could call it a 2048 x 1536 or a 3.1 Megapixel image.

You may now try the following exercises.

E12) What do you understand by the term "Brightness of Image"? How it is different from the Contrast of any Image?

So far, we have been using the word resolution and discussed it at several places. In the following section, we shall highlight resolution in detail.

1.8 IMAGE RESOLUTION

How does image resolution play out on my computer monitor? The computer screen you are looking at right now is set at a particular resolution as well. The larger the screen, the larger you likely have your screen resolution set. If you have a 17" monitor, it is likely you have it set at 800 x 600 pixels. If you have a 19" screen it is likely set at 1024 x 768. You can change the settings but these are optimum for those screen sizes.

Now, if your monitor is set to 800 x 600 and you open up an image that is 640 x 480, it will only fill up a part of your screen. If you open up an image that is 2048 x 1536 (3.1 megapixels) then you will find yourself moving the slider bar around to see all the different parts of the image. It just won't fit. Add to that the fact that the computer monitor has a finite number of pixels per inch available (like 72) so if you are going to display your image on a monitor only, you would have to reduce the quality down to 72 to save file space. If you are going to put your image on a webpage or email it to a friend then you will want to first make it a useful size. Not too big, not too small. May be 200 or 300 pixels high would be a nice size. This means that Image resolution is highly related to the size of image and its pixel density.

We learned from the last section that the term Image Resolution relates to pixel density i.e. the number of pixels in an image, which relates to clarity of image. Parameters used to determine the Resolution of any image involves the dimensions of image i.e. its width and height and the total number of pixels in the image, as well. Say, an image has following dimensions i.e. 2048 pixels wide and 1536 pixels high thus total number of pixels in the image is (2048 x 1536) i.e. 3,145,728 pixels (or 3.1 Megapixels), i.e the image has resolution of 3.1 megapixels. As the

megapixels in the pickup device in your camera increase so does the possible maximum size image you can produce. This means that a 5 megapixel camera is capable of capturing a larger image than a 3 megapixel camera.

In general the Resolution is classified as Spatial Resolution and Intensity. In simple terms, Spatial Resolution relates to the concentration of pixels in the creation of digital images and Intensity resolution of an image relates to blurredness or sharpness of an image, depending on the intensity of resolution.

- **Intensity resolution** deals with the Intensity of resolution i.e. the number of pixels per square inch, which determines the clarity or sharpness of an image. Whereas, Spatial resolution refers to the number of pixels used in making an image.
- **Spatial resolution**, in general is defined as the number of pixels per inch of an image, the term deals with the number of pixels, pixel density, and quantization levels. Thus, it depends on the sampling and quantization processes. If the spatial resolution is not satisfactory, the quality of the image would not be satisfactory. In General, if the image has more pixels, the quality of the image is supposed to be higher. but, this is not correct, because an image may have more pixels, but it may not be providing sufficient information regarding other factors such as optical resolution, the distance of an object from the camera ,the field of view etc. which plays an important role in defining image quality.

Images with a higher number of pixels per square inch are sharp and hence said to have a higher Spatial resolution i.e. Spatial resolution is the number of pixels used for the construction of a digital image. Images that have a higher spatial resolution are made of a greater number of pixels and they are much clear as compared to the images with lower spatial resolution. In radiology, spatial resolution is used to differentiate between objects located close to each other. What do we understand by ~~what~~ spatial resolution? If we have to compare two images, to see which one is more clear or which has more spatial resolution, we need to bring the two images to same size.

Since, the spatial resolution is a measure to the clarity of image, so for different devices, different measure has been made to measure it. For example Dots per inch(Dots per inch or DPI is usually used in printers), Lines per inch(Lines per inch or LPI is usually used in laser printers), Pixels per inch (Pixel per inch or PPI is measure for different devices such as tablets , Mobile phones etc.).The higher is the PPI, the higher is the quality. In order to understand that how PPI is calculated, lets calculate the PPI of a mobile phone.

First of all we will use Pythagoras theorem to calculate the diagonal resolution in pixels. It can be given as $C = \sqrt{a^2 + b^2}$, where a and b are the height and width resolutions in pixel and C is the diagonal resolution in pixels. Now we will calculate PPI
 $PPI = C / \text{diagonal size in inches}$

We shall see the following example.

Example 5: Calculate PPI or pixel density. For the sample mobile phone with the height and width resolutions in pixel as 1080 x 1920 pixels, and the diagonal size in inches of sample mobile phone is 5.0 inches.

Solution: So, putting those values in Pythagoras theorem i.e. the equation $C = \sqrt{a^2 + b^2}$ gives the result $C = 2202.90717$, Given, the diagonal size in inches of sample mobile phone is 5.0 inches, therefore $PPI = 2202.90717/5.0 = 440.58 = 441$ (approx).

That means that the pixel density of sample mobile phone is 441 PPI.

Pixel resolution, the term resolution refers to the total number of count of pixels in an digital image. For example. If an image has M rows and N columns, then its resolution can be defined as M X N. If we define resolution as the total number of pixels, then pixel resolution can be defined with set of two numbers. The first number the width of the picture, or the pixels across columns, and the second number is height of the picture, or the pixels across its height (rows). We can say that the higher is the pixel resolution, the higher is the quality of the image. We can calculate mega pixels of a camera using pixel resolution.

Pixel resolution in Megapixels = {Column pixels (width) X Row pixels (height)} / 1 Million.

Thus the size of an image can also be defined by using its pixel resolution.

Size of image = pixel resolution X bpp(bits per pixel)

Example 6: Calculate pixel resolution of a camera in mega pixels, capturing an image of dimension: 2500 X 3192.

Solution: Its pixel resolution = $2500 * 3192 = 7982350$ bytes. Dividing it by 1 million = $7.9 = 8$ mega pixel (approximately).

Now, let us discuss some important terms like *Aspect ratio*, *Dots per Inch*, *Lines per Inch*, etc. which are closely associated with the concept of image resolution

- *Aspect ratio is another important concept with the pixel resolution.* Aspect ratio is the ratio between width of an image and the height of an image. It is commonly explained as two numbers separated by a colon (8:9). This ratio differs in different images, and in different screens. The common aspect ratios are: 1.33:1, 1.37:1, 1.43:1, 1.50:1, 1.56:1, 1.66:1, 1.75:1, 1.78:1, 1.85:1, 2.00:1, etc.

Advantage of Aspect ratio is that it maintains a balance between the appearance of an image on the screen, means it maintains a ratio

between horizontal and vertical pixels. It does not let the image get distorted when aspect ratio is increased. Aspect ratio tells us many things. With the aspect ratio, you can calculate the dimensions of the image along with the size of the image.

Example 7: Given an image is an gray scale image with aspect ratio of 6:2 and pixel resolution of 480000 pixels, calculate the following:

- Resolve pixel resolution to calculate the dimensions of image
- Calculate the size of the image

Solution: Following are given:

Aspect ratio (i.e $c : r$) = 6 : 2

Pixel resolution (i.e. $r \times c$) = 480000

We know that bits per pixel for a grayscale image is 8bpp.

Here, we are required to find the number of rows(r) and the number of columns(c).

Using the Aspect ratio, we get $c : r = 6 : 2$ that is $c = 6r / 2$

Using the pixel resolution, we get $c = 480000 / r$

Comparing both relations in c and r , we get

$$\frac{6r}{2} = \frac{480000}{r} \Rightarrow r^2 = \sqrt{\frac{(480000 \times 2)}{6}} \text{ i.e. } r = 400$$

Putting $r = 400$, we get $c = 1200$;

Thus, Rows = 400 and Columns = 1200

Now we solve for the size of image.

Size of image in bits = rows * cols * bpp

Size of image in bits = $400 * 1200 * 8 = 3840000$ bits

Size of image in bytes = Size of image in bits / 8 = 480000 bytes

Size of image in kilo bytes = 469kb (approx).

- **Dots per inch(DPI):** If you have an image that is 640 x 480. How big a print can you make? Well, the true answer is you can make as big a print as you want but very quickly you will start to see "blocks" (pixelization) and the quality will drop off. To maximize the capability of your printer, you should print a picture a size that the printer can handle. Here we introduce a new term "dots per inch" (dpi) or "pixels per inch" (ppi).

The DPI is often relates to PPI (Pixels Per Inch), whereas there is a difference between these two. DPI or dots per inch is a measure of spatial resolution of printers. In case of printers, DPI means that how many dots of ink are printed per inch when an image gets printed out from the printer. Remember, it is not necessary that each Pixel per inch is printed by one dot per inch. There may be many dots per inch used for printing one pixel. The reason behind this that most of the colour printers uses CMYK model. The colours are limited. Printer has to choose from these colours to make the colour of the pixel whereas within pc, you have hundreds of thousands of colours. The higher is the dpi of the printer, the higher is the quality of the printed document or image on paper. Usually some of the laser printers have dpi of 300 and some have 600 or more.

Example 8: Determine the optimum size of the 640 X 480 image that can be printed at 200 DPI.

Solution: We have a 640 x 480 image and you want to print it at 200 dpi. 640 divided by 200 equals 3.2 and 480 divided by 200 equals 2.4 so if you print this picture at 3.2" x 2.4" you will get a print with 200 dots per inch. We recommend 200 dpi as a minimum for good quality prints.

Example 9 : Let's say we want to print an 8" X 10" picture at 300 dpi. What resolution must we have?

Solution: 300 times 8 is 2400 and 300 times 10 is 3,000. So, we would need a 3,000 x 2,400 image i.e., 7.2 megapixels.

- **Lines per inch:** As dpi refers to dots per inch, liner per inch refers to lines of dots per inch. The resolution of halftone screen is measured in lines per inch. Table 2 shows some of the lines per inch capacity of the printers.

Table 2

Printer	LPI
Screen printing	45-65 lpi
Laser printer (300 dpi)	65 lpi
Laser printer (600 dpi)	85-105 lpi
Offset Press (newsprint paper)	85 lpi
Offset Press (coated paper)	85-185 lpi

Try an exercise.

E13) What do you understand by the term “ Resolution of an Image”?
How Intensity Resolution differs from Spatial Resolution?

In the following section, we shall discuss various applications where we user digital images.

1.9 APPLICATIONS THAT USE DIGITAL IMAGES

Image processing has got wide variety of application areas, and it is widely utilized by engineers, scientists, media , fine arts professionals and many more. Nowadays it is an exciting interdisciplinary field that borrows ideas freely from many fields. Fig. 11 illustrates the relationships between image processing and other related fields.

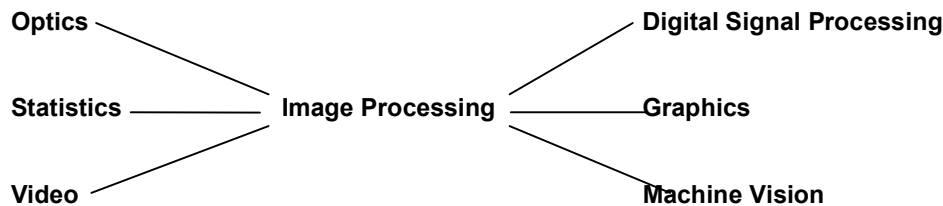


Fig. 11: Image processing and other closely related fields.

Image processing and Computer Graphics: Computer graphics and image processing are very closely related areas. Image processing deals with raster data or bitmaps, whereas computer graphics primarily deals with vector data. Raster data or bitmaps are stored in a 2D matrix form and often used to depict real images. However, vector images are composed of vectors, which represent the mathematical relationships between the objects. Vectors are lines or primitive curves that are used to describe an image. Vector graphics are often used to represent abstract, basic line drawings.

The algorithms in computer graphics often take numerical data as input and produce an image as output. However in image processing, the input is often an image. The goal of image processing is to enhance the quality of the image to assist in its interpretation. Hence, the result of image processing is often an image or the description of an image. Thus, image processing is a logical extension of computer graphics and serves as a complementary field.

Image Processing and Machine Vision : The main goal of machine vision is to interpret the image and to extract its physical, geometric, or topological properties. Thus, the output of image processing operations can be subjected to more techniques, to produce additional information for interpretation. *Artificial vision* is a vast field, with two main subfields- machine vision and computer vision. The domain of *machine vision* includes many aspects such as lighting and camera, as part of the implementation of industrial projects, since most of the applications associated with machine vision are automated visual inspection systems. The applications involving machine vision aim to inspect a large number of products and achieve improved quality controls. *Computer vision* is more ambitious. It tries to mimic the human visual system and is often associated with scene *understanding*. Most image processing algorithms produce results that can serve as the first input for machine vision algorithms.

Image Processing and Video Processing: Image processing is about still images. In fact, analog video cameras can be used to capture still images. A video can be considered as a collection of images indexed by time. Most image processing algorithms work with video readily. Thus,

video processing is an extension of image processing. In addition, images are strongly related to multimedia, as the field of multimedia broadly includes the study of audio, video images, graphics, and animation.

Image Process and Optics: Optical image processing deals with lenses, light, lighting conditions, and associated optical circuits. The study of lenses and lighting conditions has an important role in the study of image processing.

Image Processing and Statistics: Image analysis is an area that concerns the extraction and analysis of object information from the image. Imaging applications involve both simple statistics such as counting and Mensuration and complex statistics such as advanced statistical inference. So statistics play an important role in imaging applications.

You may try the following exercise.

E14) Write any three applications of digital image processing.

Now, we summarise what we have studied in the unit.

1.10 SUMMARY

We have covered the following points:

1. A digital image is defined as a 2D discrete signal that varies over the spatial coordinates x and y , and can be written mathematically as $f(x, y)$. It is also an $n \times n$ array of elements, and each element represents the sampled intensity.
2. Image acquisition can be done using various ways and the concepts required to understand the acquisition of any image.
3. The two steps of image digitization are sampling and quantization.
4. Two models for representing an image.
5. Classification of the image on the basis of various attributes of image like colour, dimension and data types, etc.
6. Various terms which will be used again and again in the course.

1.11 SOLUTIONS/ANSWERS

- E1) Image file format is an algorithm or a method used to store and display an image. Some of the frequently used file formats are JPEG, PNG, BMP, GIF, TIFF, etc.
- E2) Some of the points of difference in analogue and digital images are:

- i. Digital image's pixel value must be discrete whereas Analog image's pixel value must be continuous.
- ii. The amplitude of digital image is finite whereas the amplitude of analog image is infinite.
- iii. It is quite possible to store all the pixels of digital image whereas It is quite impossible to store all the pixels of analog image.

E3) **Analog Image Processing:** It is applied on analog signals/images and it processes only 2 D images. Examples are television, images, photographs, painting and medical images, etc.

Digital Image Processing: It is applied to digital images (a matrix of small pixels and elements). For example, colour processing, image recognition, video processing, etc.

Optical Image Processing: An optical image of 3D object is in fact its 2D projection, which is the continuous distribution of light on a 2D surface. This 2D projection holds various types of information like 3D objects that are in focus. It also includes the study of the radiation source, and other optical processes. This optical image is available in the optical form until it is converted into analog form, leading to the area of analog image processing.

E4) **Direct or Active Imaging Systems:** These are the systems where no temporary storage on film or chemical processes is required. In such systems the acquired images are in a recognizable form. We can say that in such systems the final image is a digital image. For example, human eye, digital camera, scanning confocal microscopes, etc.

Indirect Imaging Systems: In comparison to Direct imaging systems, the Indirect imaging systems involves data processing or reconstruction before producing an image for observation. In indirect imaging systems, the image is stored in a film, which is rendered observable by a chemical process. Therefore, there is a delay in the production of images from films.

E5) The bandwidth = 500 dots per inch in both directions
Therefore sample size = 1000 dots per inch [using sample theorem]

Total number of samples = $5 \times 1000 \times 8 \times 1000 = 40000000$

E6) An image may be continuous with respect to the x - and y -coordinates, and also in amplitude. Converting such an image to digital form requires that the coordinates, as well as the amplitude, be digitized.

Sampling is digitizing the coordinate values and quantization is digitizing the amplitude values.

E7) The dimensions of pixel is represented by Δx and Δy , and they are kept the same in both the horizontal and vertical directions, so that the pixels will look like square pixels. This is called pixel

aspect ratio. Sometimes the size of the pixel is very large and numbers of pixels are very less, which makes the image distracting and meaning less. This is known as pixelization error.

- E8) As per Shannon-Nyquist theorem, the sampling frequency should be greater than or equal to $2 \times f_{\max}$, where $f_{\max}$ is the highest frequency present in the image. Otherwise, the original signal cannot be reconstructed. In other words, the number of samples required is dictated by Shannon-Nyquist theorem, this sampling theorem can be stated in terms of distance d as $d \leq \frac{1}{2f_{\max}}$.

Alternatively, it should be less than or equal to the smallest detail that is present in the image.

- E9) **Pixel model:** In this model, the RGB optical sensors map to a vector of discrete intensities which are integers. Let $p: R \times G \times B \rightarrow Z \times Z \times Z$ defined by $p(r, g, b) = (zR, zG, zB)$, where r, g, b are real valued colour intensities and zR, zG, zB are integer-valued colour intensities in the range from 0 to 255. In this case, a pixel is represented by a 1×3 column vector used to "paint" a tiny square in an image display.

Vector model: In this model, the RGB optical sensor values are mapped to vectors and the displayed image conforms to a form of triangulated image space instead of a collection of pixels. In this model, a vector image is a collection of vertices and edges. The end result of vector images is the formation of a basis for applications such as Adobe Photoshop and various video games.

- E10) The classification of an image can be done on the basis of

- i) Attributes
- ii) Colour
- iii) Dimensions
- iv) Data type.

- E11) **Monochrome images** are the images where the colour component is absent, they are further classified as Grey scale and binary images.

True colour (or full colour) images are the images where the pixel has a colour that is obtained by mixing the primary colours red, green, and blue. Each colour component is represented like a grey scale image using eight bits. Mostly, true colour images use 24 bits to represent all the colour. Hence true colour image can be considered as three-band images. The number of colours that is possible is 256^3 (i.e. $256 \times 256 \times 256 = 1,67,77,216$ colours).

- E12) **Brightness:** It refers to the average pixel intensity of the image. Similar to contrast, the image should be reasonably bright to show all the information to the viewer. However, excessive brightness may affect the quality of the image. An image may have a higher brightness but if the intensity is not optimal, the image again will not exhibit good quality. *Brightness resolution* is defined as the number of identified distinct colours and it is also called as *colour resolution*.

We know, Contrast of an image relates to recording of the differences in the magnitude of the intensity at the surface of an object. So, Contrast is the difference between maximum and minimum pixel intensities in an image. Consider two images A & B having pixel intensities between 30 to 200 and 70 to 130, respectively. Then A has more contrast than B. Again contrast is also relative. Contrast can be simply explained as the difference between maximum and minimum pixel intensity in an image. For example, consider an image, whose maximum and minimum value of intensity is 100 i.e. same then contrast will be zero because, $\text{Contrast} = \text{maximum pixel intensity} - \text{minimum pixel intensity} = 100 - 100 = 0$.

- E13) **Pixel resolution**, the term resolution refers to the total number of count of pixels in an digital image. For example. If an image has M rows and N columns, then its resolution can be defined as $M \times N$. If we define resolution as the total number of pixels, then pixel resolution can be defined with set of two numbers.
- E14) Refer Section 1.9 or Browse information over relevant websites.



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